Hall Ticket Number:

MR-21

Code: B415CB5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July 2025

Software Engineering

(Computer Science and Business System)

Time: 3 Hours

Answer ALL Questions

Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is software engineering? What are its applications?	2	2	1	2
1. b)	Differentiate between software process and project.	2	3	1	2
1. c)	How do non-functional requirements differ from functional requirements?	2	3	2	2
1. d)	Define context models in system modeling.	2	1	2	1
1. e)	How would you apply UML class diagrams in software design?	2	3	3	3
1. f)	What is the purpose of architectural styles in software design?	2	2	3	2
1. g)	What is the difference between black-box testing and white-box testing?	2	2	4	2
1. h)	What is the role of debugging in the software testing process?	2	1	4	2
1. i)	Justify the use of the ISO 9000 quality standards as a benchmark for global software quality assurance practices.	2	5	5	7
1. j)	List the key components of the RMMM plan.	2	1	5	2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	What is waterfall process model and discuss about its reasons for failure?	5	2	1	2
2. b)	List and explain various characteristics of software.	5	2	1	2
	OR				
2. c)	Describe in detail about personal and team process models.	5	2	1	2
2. d)	Explain unified process and elaborate on the unified process work products.	5	2	1	2
3. a)	Write the difficulty process for Eliciting and understanding requirements from system stakeholders.	5	2	2	2
3. b)	Explain how behavioral models help in understanding system functionality.	5	3	2	2
	OR				
3. c)	Discuss the significance of object models in system modeling. Provide an example.	5	2	2	2
3. d)	Why is it important to validate software requirements, and what are common techniques used for validation?	5	2	2	2
4. a)	How would you use UML diagrams to model both the static and dynamic aspects of a software system? Provide examples.	5	3	3	3
4. b)	Explain how sequence diagrams and collaboration diagrams can be used together to model object interactions in a system.	5	3	3	3
	OR				
4. c)	Illustrate design process to ensure design quality in software development.	5	2	3	3
4. d)	How can component diagrams be applied to represent the physical structure of a system? Provide a relevant example.	5	3	3	3

P.T.O.

5. a)	How would you evaluate different test strategies for conventional software? Provide examples of when to apply each strategy.	5	1	4	2
5. b)	Analyze the role of metrics in the design model for ensuring design quality in a software project.	5	4	4	2
	OR				
5. c)	How would you assess the importance of product metrics in software maintenance, and which metrics are most critical for this phase?	5	4	4	2
5. d)	Explain how validation testing differs from system testing, and analyze their roles in ensuring software quality.	5	2	4	2
6. a)	What is the role of software reviews and formal technical reviews in the software development process? Explain.	5	2	5	2
6. b)	Describe the process of risk identification and how it helps in creating a proactive risk management strategy.	5	2	5	2
	OR				
6. c)	Compare and contrast reactive and proactive risk strategies in risk management.	5	4	5	2
6. d)	How would you determine the success of software reliability measures in meeting customer expectations for a software product?	5	5	5	7

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome